

PAPER_946: Merger Lagrangian Variation

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** `smbh_binary_mergers.py` (MergerLagrangianVariation) **Calculator:** MergerLagrangianVariationCalc (CP4 #530) **CVW:** v2.0.0 compliant

Abstract

We derive the UQFF Euler-Lagrange equation for SMBH binary merger phonon fields. The stationarity condition $\delta S / \delta \varphi_{\text{merger}} = 0$ yields a critical radius r_{crit} at which the gravitational Lagrangian $\mathcal{L}_{\text{grav}} = -\beta_i \sum_i U_{g,i} \cdot M / d_g \cdot [\text{UA}]$ balances the phonon Lagrangian $\mathcal{L}_{\text{phonon}} = F_n \cdot \Phi$. The 26-channel buoyancy sum governs the transition.

1. Merger Lagrangian

$$\mathcal{L}_{\text{merger}} = \mathcal{L}_{\text{grav}} + \mathcal{L}_{\text{phonon}}$$

$$\mathcal{L}_{\text{grav}} = -\beta_i \sum_{i=1}^{26} U_{g,i} \cdot \frac{M}{d_g} \cdot [\text{UA}]_i$$

$$\mathcal{L}_{\text{phonon}} = F_n \cdot \Phi$$

2. Stationarity Condition

$$\frac{\delta S}{\delta \varphi_{\text{merger}}} = \frac{\partial}{\partial E_{\text{net}}} \left(-\beta_i \sum U_{g,i} \cdot \Omega_g \cdot \frac{M}{d_g} \cdot [\text{UA}] + F_n \cdot \Phi \right) = 0$$

This yields the critical radius:

$$r_{\text{crit}} = \frac{2\beta_i \sum U_{g,i} \cdot M}{|F_n \cdot \Phi|}$$

3. Physical Interpretation

Below r_{crit} , gravitational effects dominate, and the merger phonon field acts as a passive tracer. Above r_{crit} , phonon back-reaction becomes significant and modifies the inspiral dynamics through buoyancy feedback.

4. Source Data

- **File:** `smbh_binary_mergers.py`
 - **Session:** 213
 - **CP4 Class:** MergerLagrangianVariationCalc (#530)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. Arnowitt, R., Deser, S. & Misner, C.W. (1962) - Gravitation (Wiley)